

VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY

UNIVERSITY OF TECHNOLOGY

FACULTY OF COMPUTER SCIENCE AND ENGINEERING



💡 **Course: Machine Learning - 242** 💡

Assignment 2

💡 **Class: CC01 - Group LHPD2** 💡

Team Members, Team Repository Link and Team Report

Our repository is at this link: https://github.com/pdz1804/ML_LHPD2.

This README that we submitted to you is stored in the repository at [ML_LHPD2/notebooks/assignment2/ML_LHPD2_Ass2_README.md](#).

Also, our team's report which contains detailed information (data analysis, data preprocessing, normalization, models training process and evaluation of those trained models) for this Assignment 1 is stored in the repository at [ML_LHPD2/reports/final_project/](#).

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I. Group Project Requirements

Project Organization

Group Formation:

- The group consists of **5 members**.
- **Nguyen Quang Phu** is the **repository owner**.

Collaboration Workflow:

- The **main repository** is managed by the designated team member.
- Other members **fork** the repository and contribute via **pull requests**.
- **Roles and responsibilities** are clearly defined for each team member.

GitHub Repository Structure

We are implementing **ALL machine learning models from the syllabus** using:

- **ONE unified dataset** with one use-case before the midterm (Chapters 2-6) (which we do in this Assignment - Assignment 1).
- **THAT dataset** for that same use-case for the second stage after midterm (Chapters 6-10) (which we will do in our Assignment 2).

Repository Setup:

- The **main repository** is created by the team lead.
- Team members **fork** and contribute via **pull requests**.
- **Branching strategy** is used for different models and features.
- **Comprehensive documentation** is maintained.

Suggested Repository Structure:

```
📁 Project-Root
|— 📁 data/           # Raw & Processed Datasets
|— 📁 notebooks/     # Jupyter Notebooks for EDA & Preprocessing
```

```
|— 📁 models/           # ML Model Implementations
|— 📁 other/            # Some other related things here
|— 📁 reports/          # Contain figures and Reports
|— 📁 src/              # Contain src code of how to process data, build features
|— 📁 tests/            # Evaluation Results & Comparisons
|— 📄 README.md         # Project Documentation
|— 📄 requirements.txt  # Dependencies
```

3 Key Requirements

- ✓ Well-documented problem statements & model variations
 - ✓ Consistent structure for model implementations
 - ✓ Comprehensive testing of all components
 - ✓ Comparative analysis of models
 - ✓ Code reviews and pull requests
 - ✓ Version control best practices
-

4 Each Team Member Should:

- **Actively contribute to the repository** 🚀
 - **Document their work thoroughly** 📝
 - **Review and improve others' code** 👁️
 - **Participate in group discussions** 💬
 - **Help maintain code quality** ✓
-

II. Completed Tasks

1 Data Collection & Preprocessing

We have **collected and preprocessed** the dataset, and it has been uploaded to **Kaggle**.

🔗 **Kaggle Dataset:** [Tweets Clean PosNeg v1](#)

📌 **Dataset Preview:**

NGUYEN QUANG PHU · UPDATED 16 DAYS AGO

5

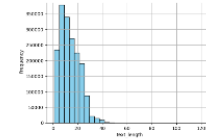
<> Code

Download

⋮

tweets_clean_posneg_v1

Merged & cleaned dataset from sentiment140, twitter-tweets, & twitter-sentiment



Data Card Code (1) Discussion (0) Suggestions (0) Settings

About Dataset

This dataset is created by merging and cleaning three sentiment analysis datasets:

- **sentiment140** (kazanov/sentiment140)
- **twitter-tweets-sentiment-dataset** (yasserh/twitter-tweets-sentiment-dataset)
- **twitter-sentiment-dataset** (saurabhshahane/twitter-sentiment-dataset)

The resulting dataset consists of **1,628,166 entries** with the following columns:

- **target**: Sentiment label of the text (numeric representation: 0 for negative, 4 for positive).
- **text**: Original text from the corpus.
- **text_clean**: The text after cleaning (removal of noise, special characters, etc.).
- **text_length**: Length of the original text (number of words).
- **text_clean_length**: Length of the cleaned text (number of words).

This dataset is ideal for sentiment analysis and text classification tasks.

Edit

Usability ⓘ

10.00

License

[MIT](#) [Edit](#)

Expected update frequency

Annually [Edit](#)

Tags

Online Communities

Text

NLP

Feature Engineering

✎

✓ Tasks Completed:

- Collected **raw tweets dataset** 📄
- Applied **data cleaning & preprocessing** ✎
- Uploaded the final dataset to **Kaggle** 📁

2 Models that we have done in Assignment 1

- **Train and Evaluate Models** 🚀
 - Decision Tree
 - Random Forest
 - XGBoost
 - Logistic Regression
 - Naive Bayes
 - GA, HMM, Bayesian Network
 - MLPClassifier and simple Perceptron (ANN)
 - CNN / LSTM for deep learning
- **Compare Model Performance** 🏆
- **Create Visualizations for Analysis** 📊

- **Write Final Report & Documentation** 📝

3 Models that we have done in Assignment 2

- **Train and Evaluate Models** 🚀
 - Graphical Models (Bayesian Networks, HMM)
 - Support Vector Machines (kernel and regularization)
 - PCA for Dimension Reduction
 - Feature Selection Strategies
 - Topic Modeling, Variance, Chi-squared
 - LDA as linear classifier
 - Ensemble Models / Methods
 - Random Forest
 - XGBoost
 - Stacking: Stack multiple Logistic Regressions; Stack multiple "good" models from Assignment 1
 - Voting: Use multiple Logistic Regressions and Use multiple "good" models from Assignment 1 for evaluating
 - Discriminative Models:
 - Logistic Regression
 - CRF: specifically we use BiLSTM-CRF to make use of the superior performance of BiLSTM model.

- **Compare Model Performance** 🏆

- **Create Visualizations for Analysis** 📊

- **Write Final Report & Documentation** 📝

💡 IV. Contributions and Task Distribution

Task Distribution for Assignment 1

Team Member	Task

Nguyen Quang Phu	Team leader; Repository management; Participate and Ensure everything stays on schedule and verify all work done by other members. 📁
Pham Huynh Bao Dai	Data preparation; Data preprocessing; feature engineering; Visualization; Document Data Collecting; Preprocessing and Merging. ⚙️
Nguyen Thanh Dat	Consistent model training and evaluating; Model implementation (Decision Tree, Random Forest, XGBoost, Perceptron - ANN, MLP); Document Model Implementation. 🌳
Nguyen Tien Hung	Consistent model training and evaluating; Model implementation (GA, HMM Bayesian Network, Logistic Regression, LSTM); Document Model Implementation. 📈
Nguyen Thien Loc	Model evaluation, hyperparameter tuning, Model Comparison, Document Performance analysis. 🔍

Task Distribution for Assignment 2

Team Member	Task
Nguyen Quang Phu	Team leader; Manage repository and schedule; Stacking & Voting with Logistic Regressions; Final integration & verification. 📁
Pham Huynh Bao Dai	Implement PCA, SVM; Visualizations; Document his Models' workflow ⚙️
Nguyen Thanh Dat	Implement LDA, Random Forest, XGBoost; Document Ensemble Methods and Classifiers. 🌳
Nguyen Tien Hung	Build Bayesian Networks, HMMs, Logistic Regression; Stacking with good models from Assignment 1; Document Graphical Models. 📈
Nguyen Thien Loc	Implement BiLSTM-CRF; Evaluate models; Voting with good models; Performance Analysis and Comparison. 🔍

📌 V. How to Run This Project

1 Clone the Repository

```
git clone https://github.com/pdz1804/ML_LHPD2
```

2 Setting Up the Environment

If you want to train the model in local (on your computer, in VSC...), to install the necessary dependencies, refer to the environment file (`environment.yml`) provided in the repository. Ensure you have Conda installed to create and activate the required environment.

```
conda env create -f environment.yml
conda activate ml_env # Replace ml_env with your environment name
```

3 Explore the Project

- First, you should revisit the Project structure above to see what is the use of those folders and subfolders in this project.
- In details:
 - To know how we collect all the data, visit the `data` folder.
 - To see the final preprocessing dataset, visiting this link: [Tweets Clean PosNeg v1](#)
 - To know how we preprocessing, make all the datasets consistent and how do we merge them together, visit the location `src/data/process.ipynb` and its utility file `src/data/preprocess.py`
 - To know how and what we use to make the features for all the texts/documents in the dataset, visit the location `src/features/build_features_utils.py`
 - To know how we perform hyperparameter tuning, do k-fold cross-validation and train our models, visit `src/models/models_utils.py` and `src/models/train_model.ipynb`
 - We have also created some visualization functions inside the location `src/visualization/` . Although now we have not used it anywhere in code because all the visualization or plotting now we just use the **in-line-function-plotting**, but we intend to use it later in Assignment 2.
 - The `environment.yml` contains the information about the libraries or dependencies that we use when performing and executing our project.



VI. License

This project is licensed under the **MIT License**.



VII. Contact

For any questions or contributions, please contact:

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🔗 GitHub: <https://github.com/pdz1804/>

✨💖 Contributors 💖✨

💖 We fairly contribute to this repository with dedication and teamwork! 💖

